

Data Analysis of 5G Packets

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Section 1

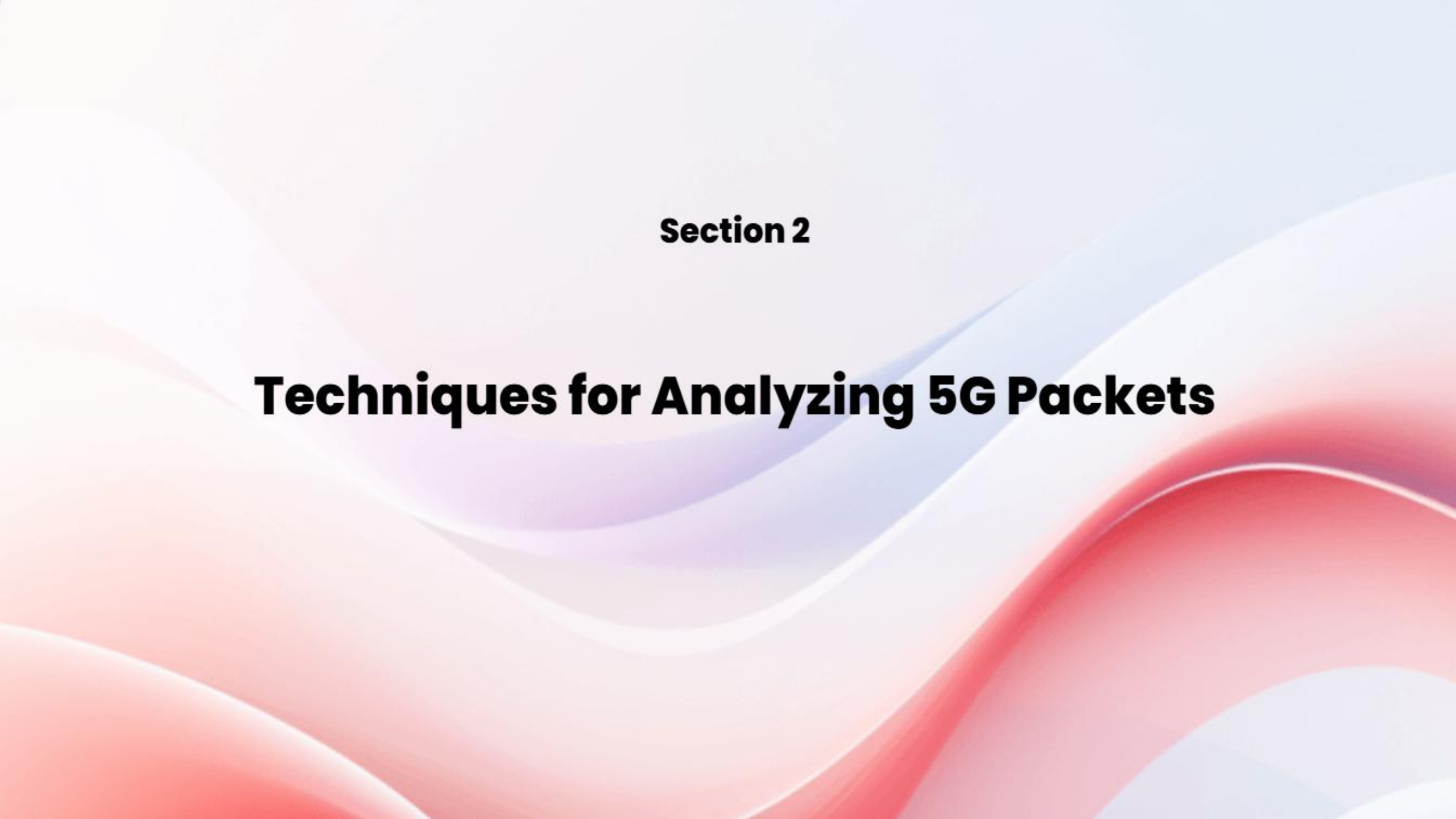
Understanding 5G Packet Data

Overview of 5G Network Architecture

STRUCTURE OF DATA TRANSMISSION

- Data is taken from User equipment.
- U.E data is transmitted to Base Station in wireless mode.
- Then from Base Station to last node through connecting wires.
- So, the data is routed through different nodes and arrives at the final destination.





Section 2

Techniques for Analyzing 5G Packets

Extracting Dictionaries from the given file

```
sn: 3170  
entry_time: 1730544532.138246  
exit_time: 1730544532.157495  
length_pkt: 128  
ID_pkt: 36822
```

```
sn: 3171  
entry_time: 1730544532.18832  
exit_time: 1730544532.207435  
length_pkt: 128  
ID_pkt: 36821
```

```
sn: 3172  
entry_time: 1730544532.238374  
exit_time: 1730544532.257498  
length_pkt: 128  
ID_pkt: 36820
```

Extracting Entry and Exit Time

Python

```
Entry Times: [1730544347.449835, 1730544351.481821, 1730544359.421818, 1730544374.181447, 1730544374.208595, 1730544374.208
```

```
Exit Times: [1730544347.469965, 1730544351.504944, 1730544359.445, 1730544374.20503, 1730544374.227611, 1730544374.227615,
```

Key Findings from 5G Packet Data Analysis

- *delay variable and delay=list1-list2*

```
Delays:
serial_no: 1
delay: 0.020129919052124023

serial_no: 2
delay: 0.02312302589416504

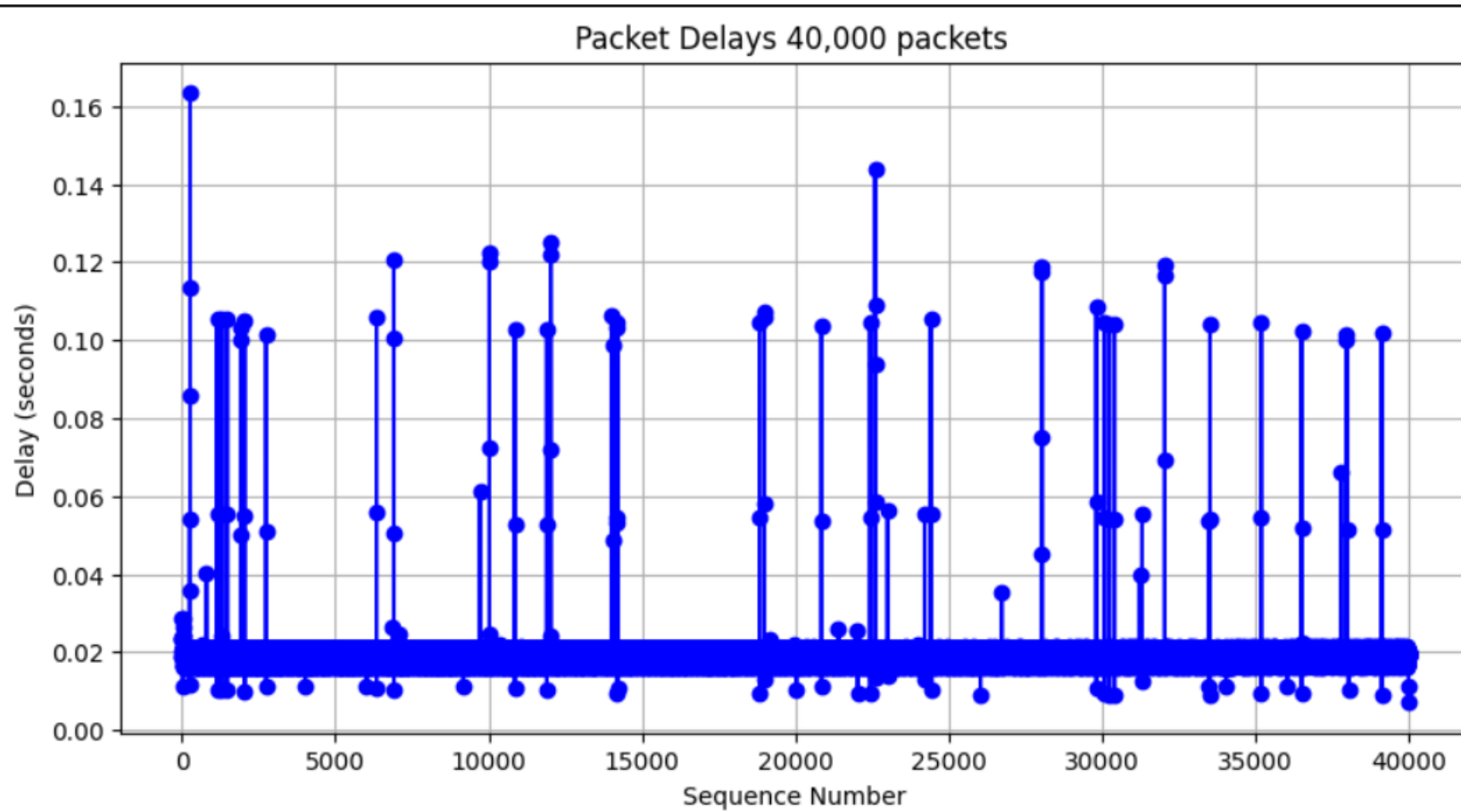
serial_no: 3
delay: 0.023181915283203125

serial_no: 4
delay: 0.023582935333251953

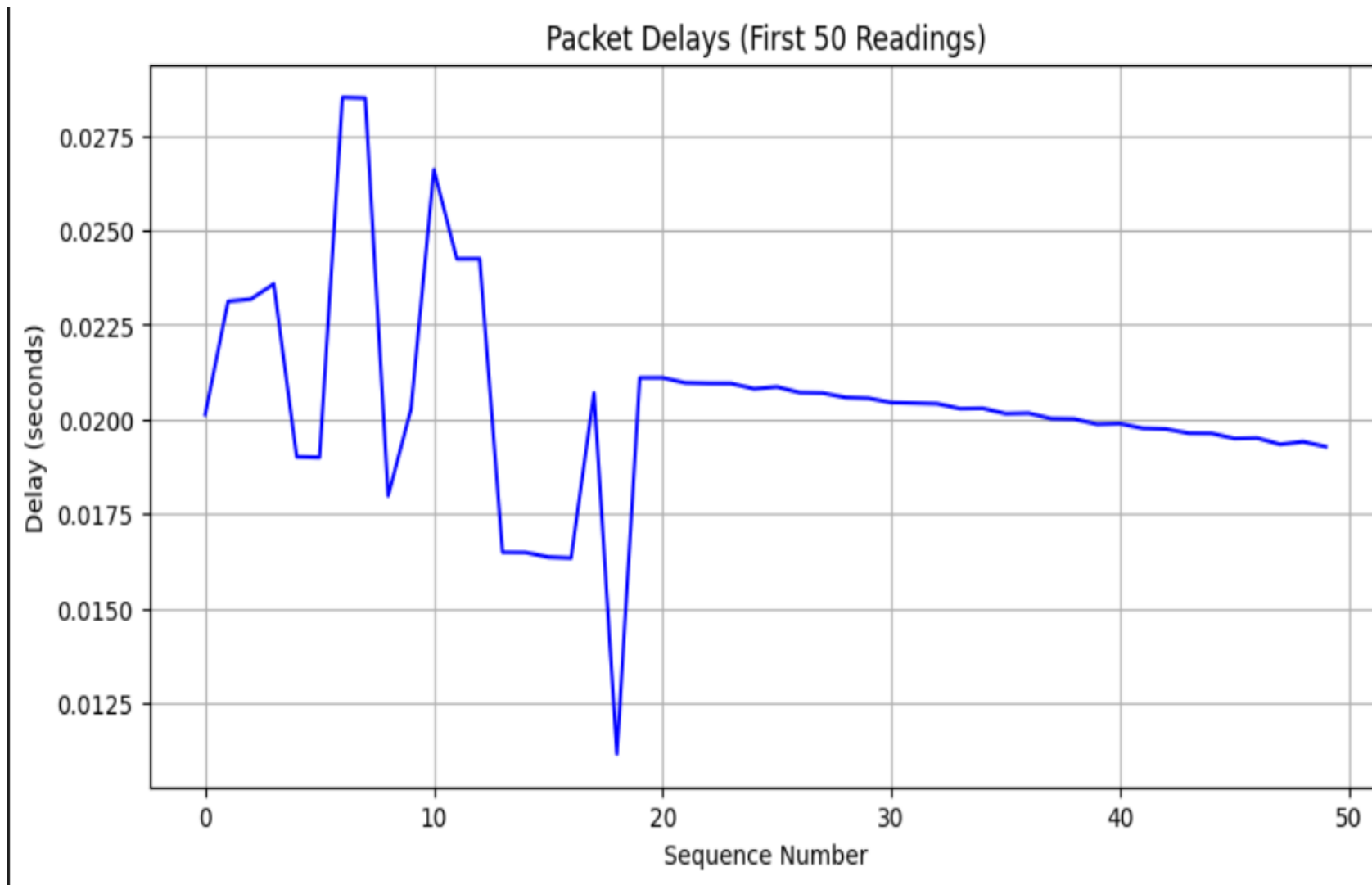
serial_no: 5
delay: 0.019016027450561523

serial_no: 6
delay: 0.01900005340576172
```

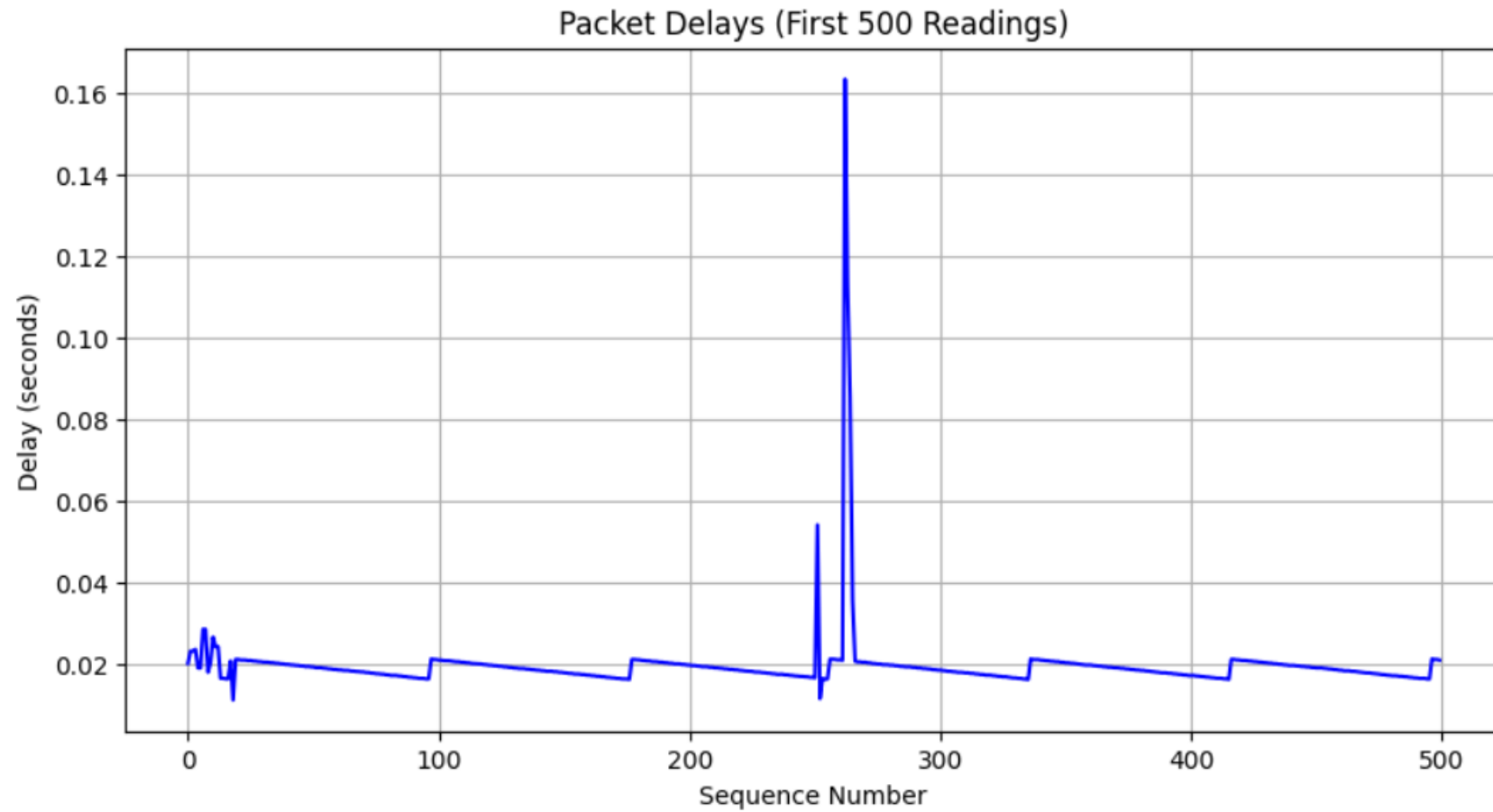

Plot of graph with sequence numbers on the x-axis and delay values on the y-axis using Matplotlib (40,000 Packets)



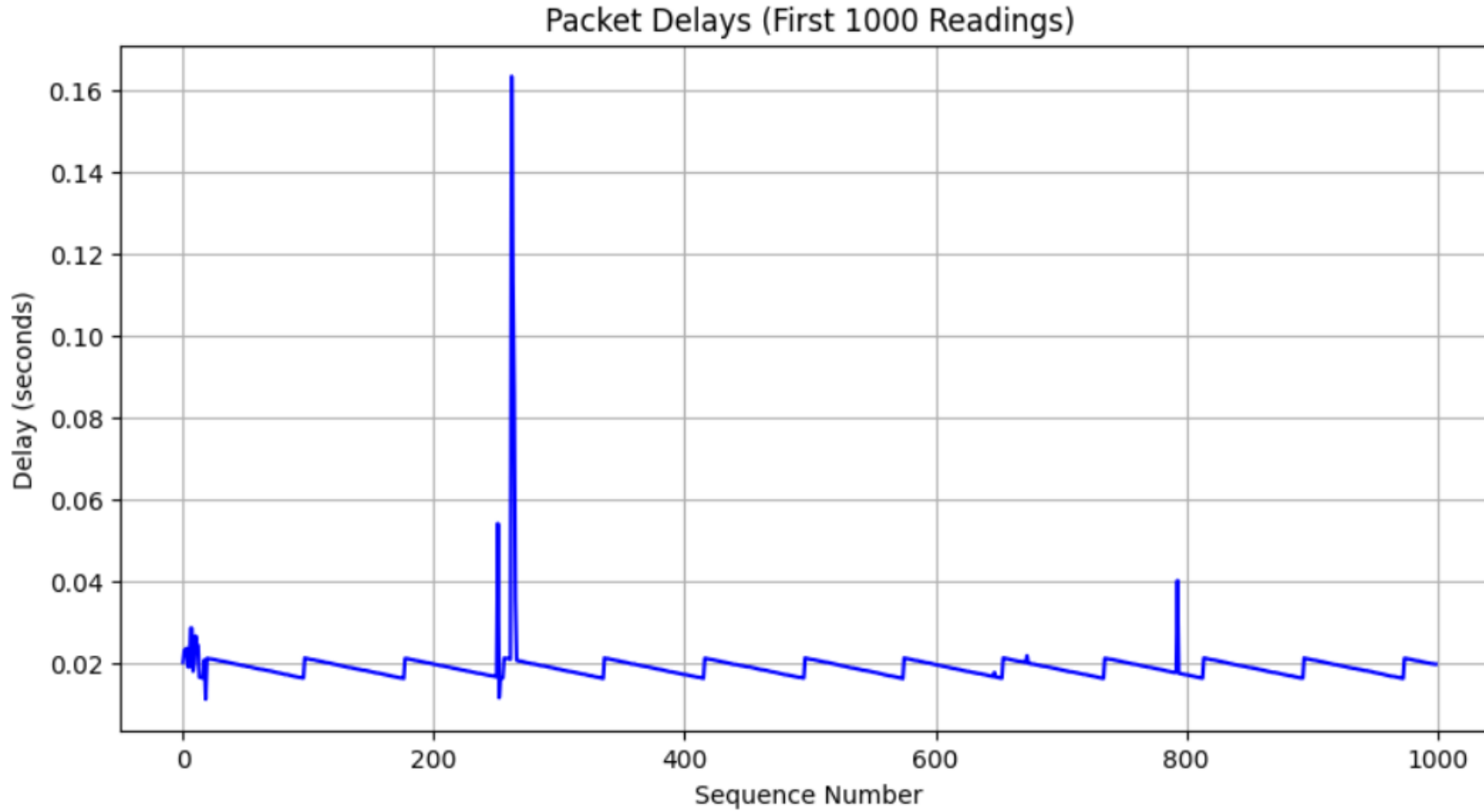
Plot of 50 readings



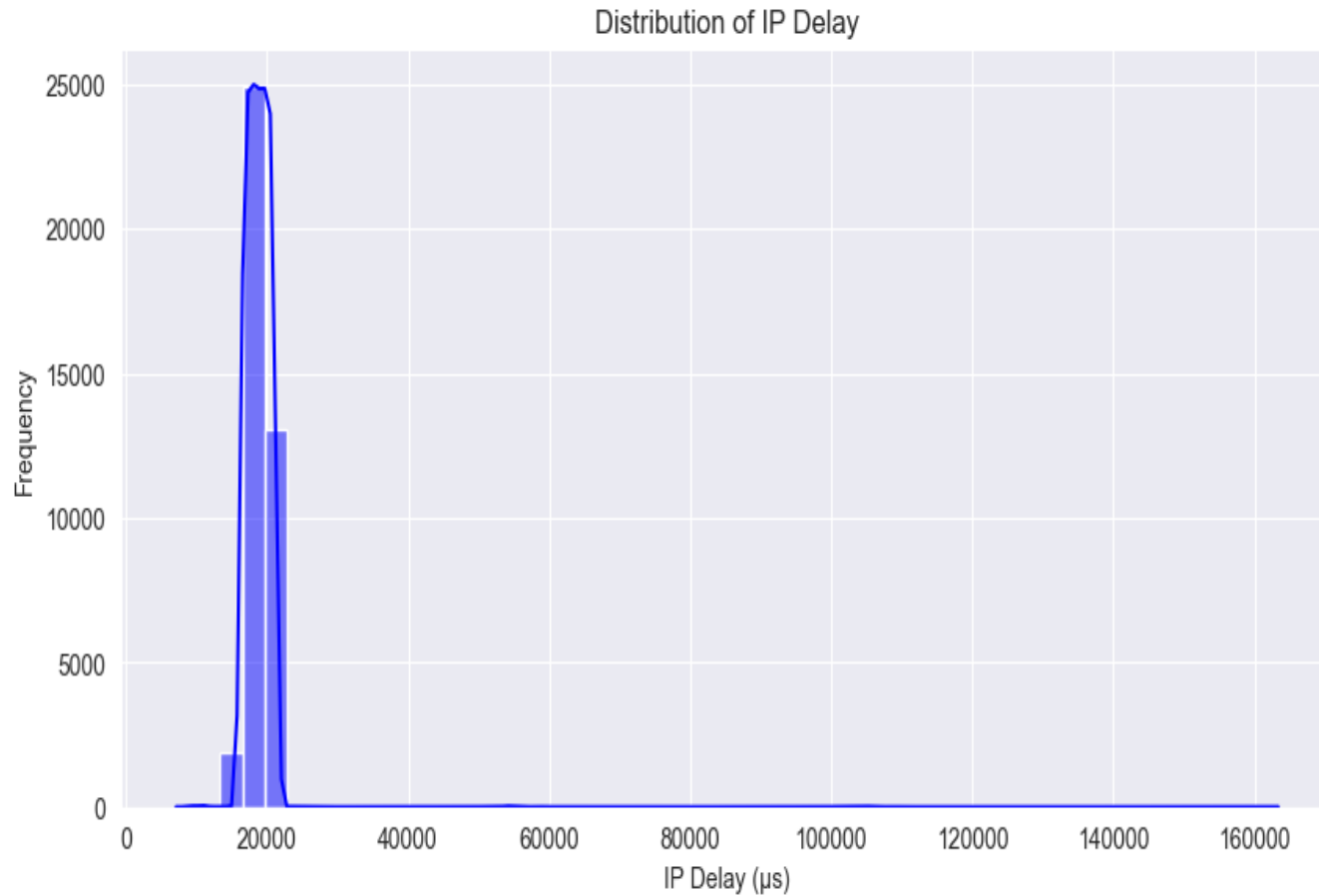
Plot of 500 readings



Plot of 1000 readings



DISTRIBUTION OF IP DELAY



```
... ♦ Maximum Delay: 163372.039795 μs
    ♦ Average Delay: 18878.679087 μs
count      39993.000000
mean       18878.679087
std        3626.670916
min        7102.966309
25%        17493.009567
50%        18744.945526
75%        20001.173019
max        163372.039795
Name: IP Delay, dtype: float64
```

FINDINGS FROM GRAPH

- KEY Findings -> Average delay is 0.018878 seconds
Highest delay is 0.163372 seconds
- Delay is inversely proportional to transmission speed of packet in the medium.

POSSIBLE REASONS FOR DELAY

- Transmission delay
- Processing and Congestion Delay
- Queuing Delay
- Propagation Delay
- Packet Loss

Plot of IP Delay and Packet Length

What does this plot show?

- The x-axis represents packet size (length in bytes), while the y-axis shows delay.
- It helps determine if larger packets take longer to transmit.

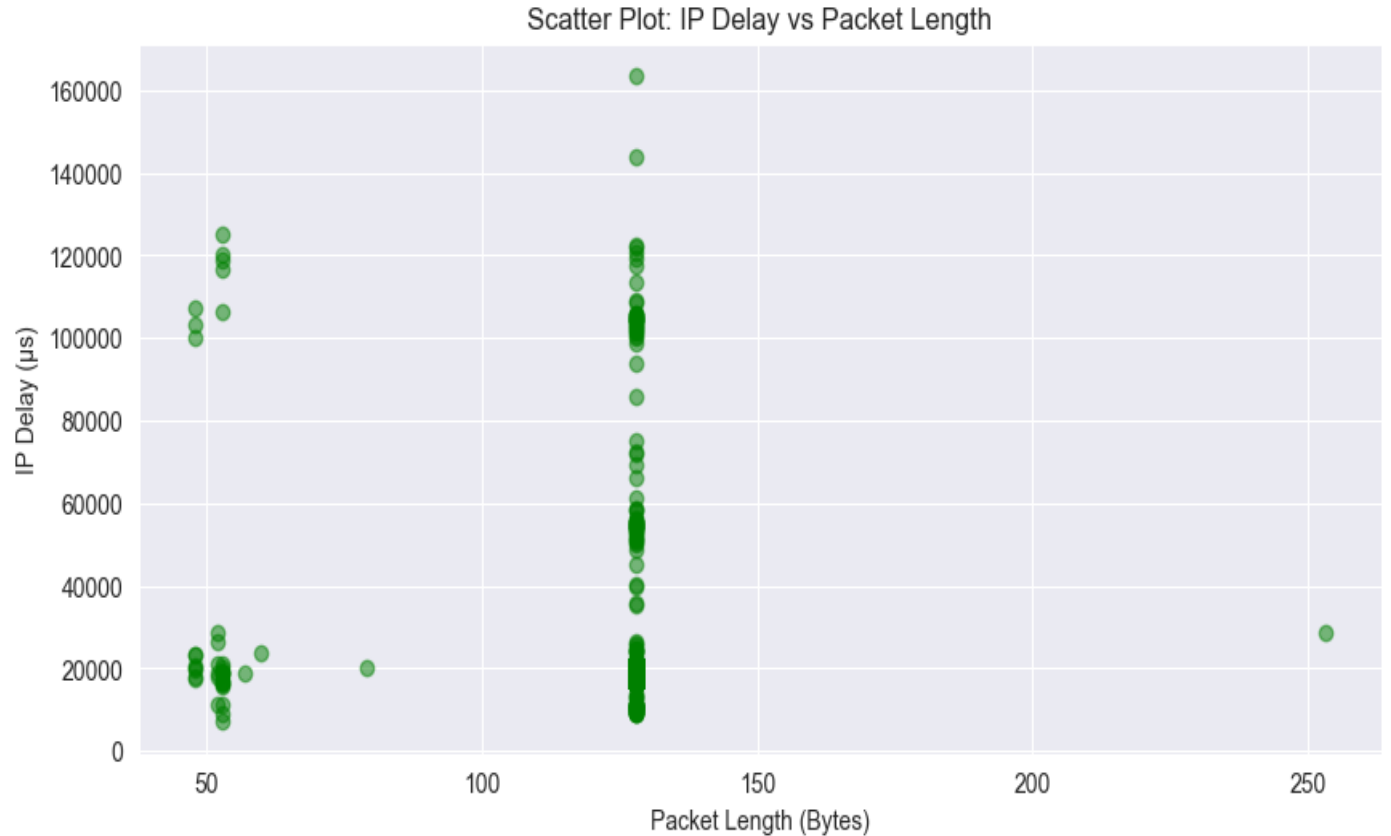
Insights

•Direct Relationship:

If larger packets have higher delay, then network transmission time depends on packet size.

If the delay remains constant, packet size significantly.

- Outliers:** Very large or small does not impact latency packets experiencing extreme delay may indicate network inefficiencies.



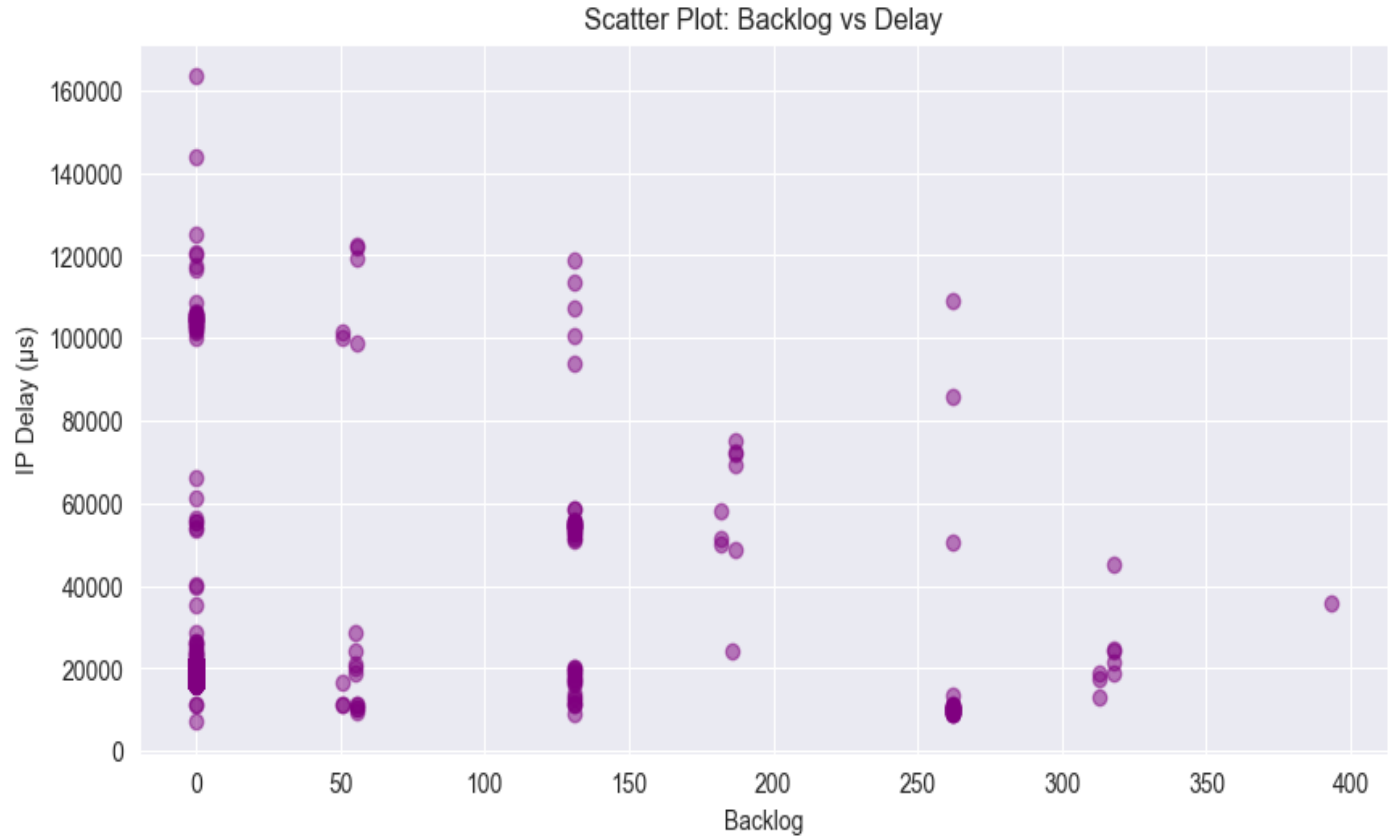
Plot of IP Delay and Backlog

What does this plot show?

- The x-axis represents backlog (queued packets waiting for transmission).
- The y-axis represents packet delay.

Insights:

- Higher backlog = Higher delay:
If there is a strong correlation, the network queue causes increased latency.
If no correlation is found, other factors like network interference may be causing delay.
- Threshold Detection: Find backlog levels where delays start increasing sharply, helping optimize network buffer sizes.



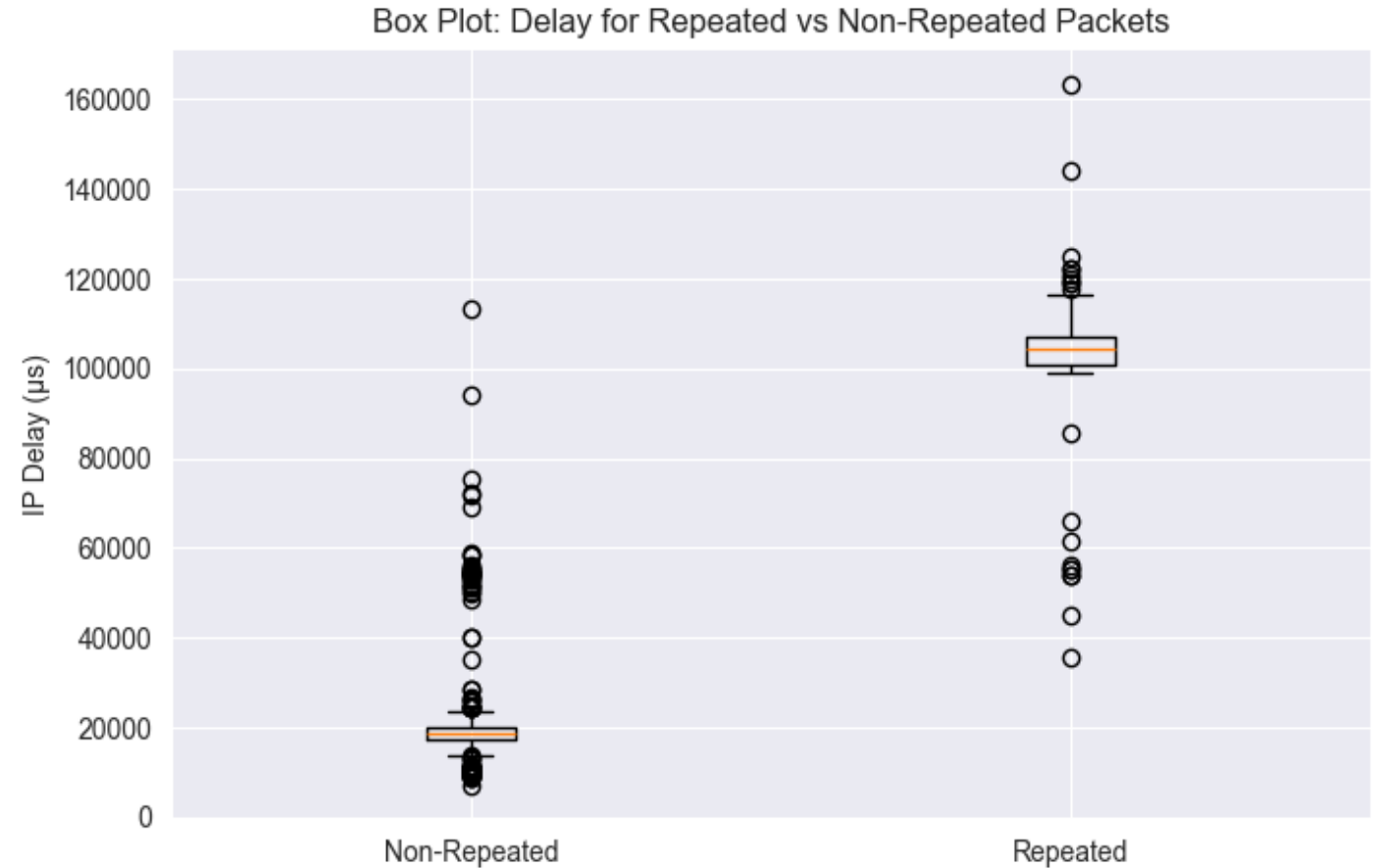
Plot of IP Delay for Repeated and Non-Repeated Packets

What does this plot show?

- This compares the delay distribution between normal and retransmitted packets.
- Box plots show median, quartiles, and outliers.

Insights:

- Higher median delay for retransmitted packets:
 - Suggests that packet retransmissions significantly increase network latency.
 - Indicates a need for better retransmission strategies (e.g., adaptive retransmissions).
- Outliers: Extreme delays in retransmitted packets may indicate inefficient retry mechanisms.





Section 3

Insights and Future Directions

Insights from the Data Analysis

After visualizing and analyzing the IP Delay, here are some key takeaways:

1) Delay Distribution Insights

If the histogram of delay is skewed right, it indicates some packets experience high delay, which might be caused by network congestion or resource allocation issues.

If there are multiple peaks, it suggests different traffic types in the system (e.g., control & data packets).

2) Delay vs. Sequence Number (SN)

If the scatter plot shows an increasing trend, it suggests that delays accumulate over time, indicating buffering, congestion, or queuing issues in the 5G system.

If certain SNs show high spikes, they could be outliers due to retransmissions or resource shortages.

3) Delay vs. Packet Length

If larger packets tend to have higher delays, then the system may struggle to handle large-sized data efficiently.

This can be mitigated using packet segmentation techniques.

4) Backlog vs. Delay

If a high backlog correlates with high delay, then network buffers may be getting full, leading to queue buildup.

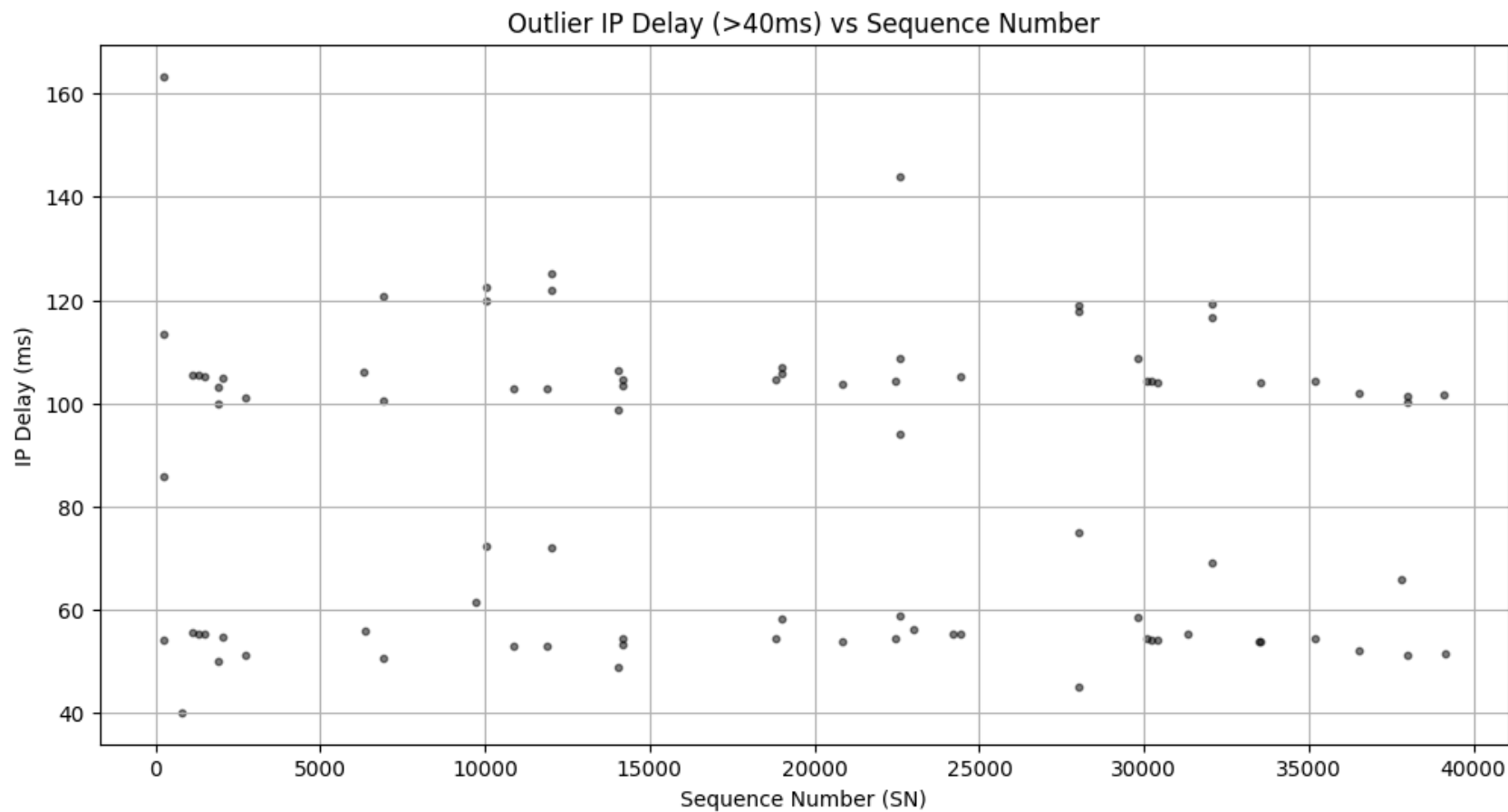
This suggests the need for optimized scheduling or queue management strategies.

5) Max & Avg Delay Insights

If the maximum delay is significantly higher than the average, it indicates potential outliers or network anomalies.

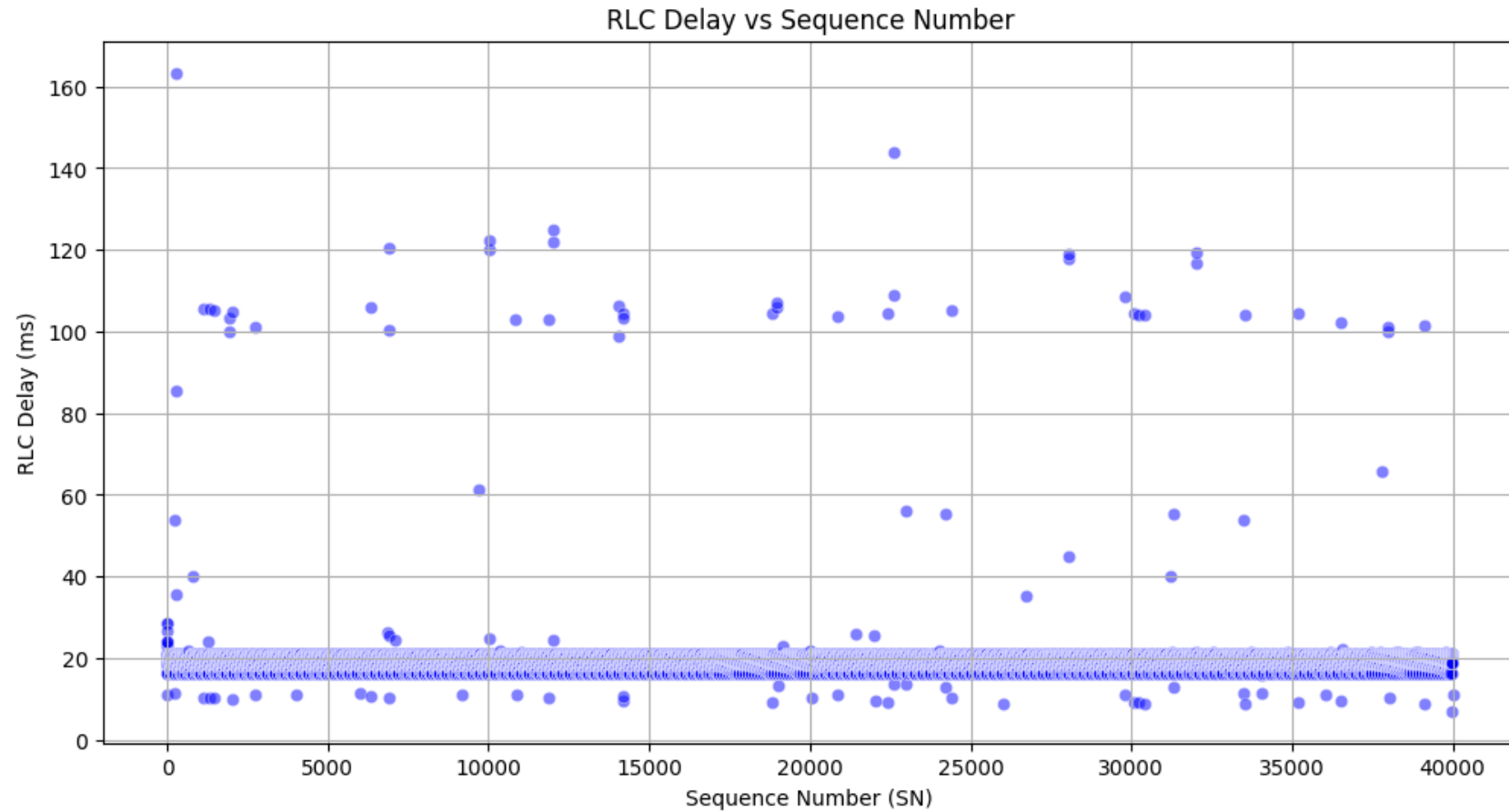
If the average delay is increasing over time, the network may be overloaded or experiencing inefficient resource allocation.

Plot of Outliers in IP Delay (>40ms)

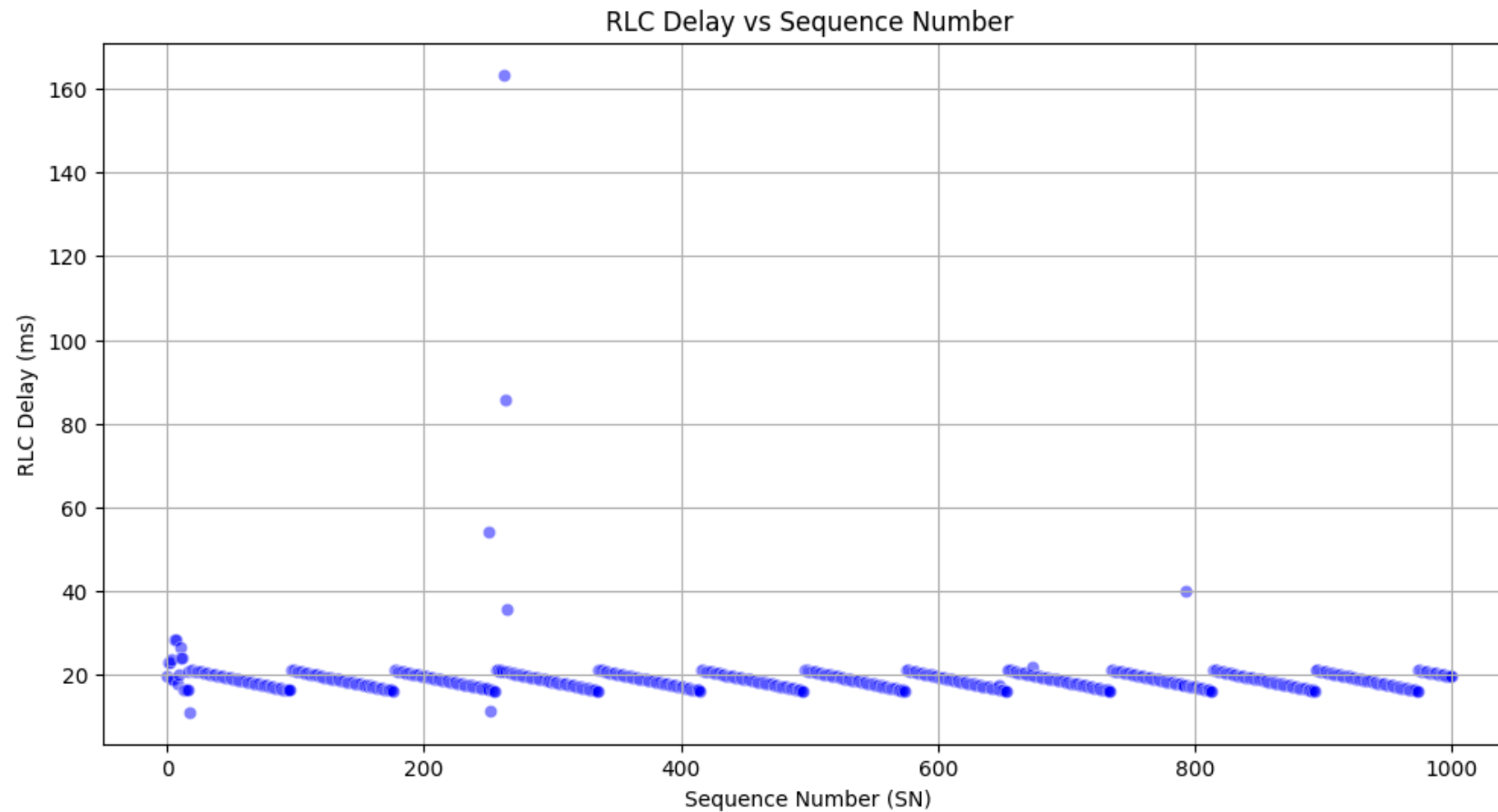


ANALYSIS OF DELAY IN RLC

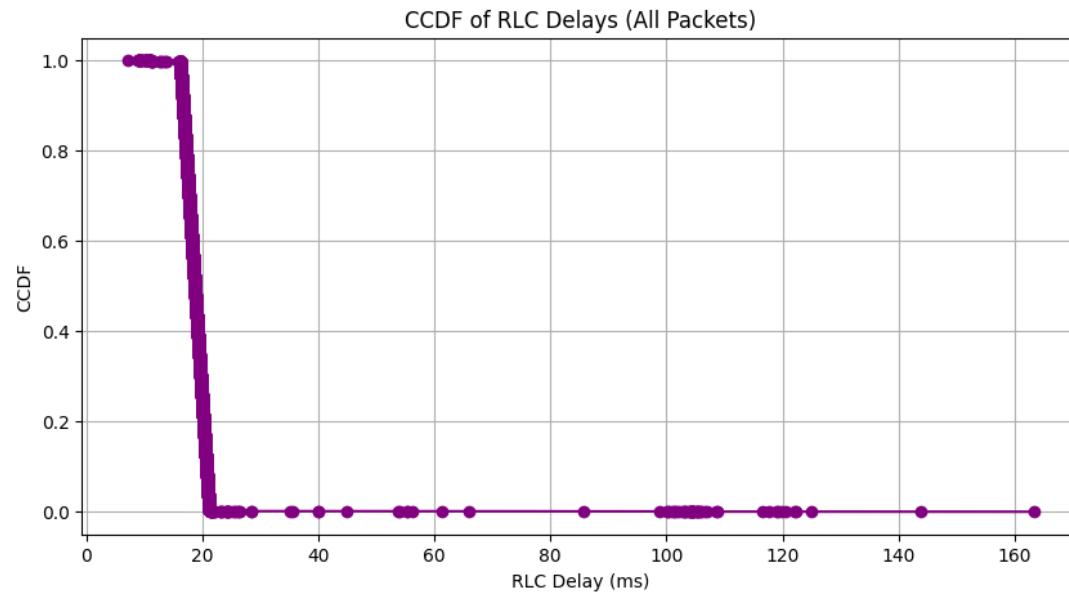
PLOT OF RLC DELAY vs SN



PLOT OF RLC Delay vs SN for first 1000 packets



Statistics of RLC Delay for all the packets



****RLC Delay Statistics****

Mean Delay: 18.829 ms

Median Delay: 18.727 ms

Min Delay: 7.082 ms

Max Delay: 163.350 ms

Standard Deviation: 3.412 ms

****Percentiles****

99.2th Percentile: 21.216 ms

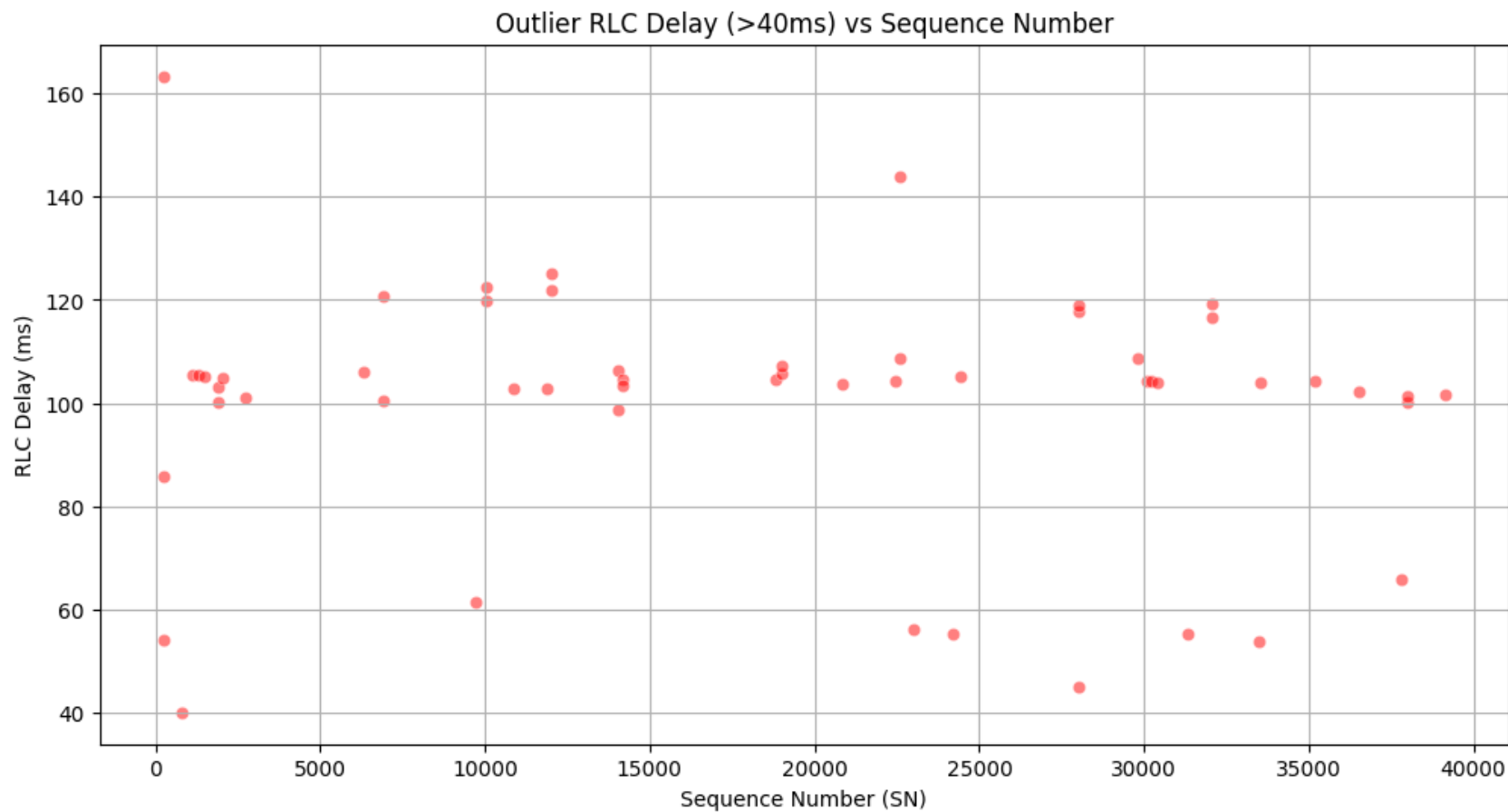
99.4th Percentile: 21.233 ms

99.6 Percentile: 21.264 ms

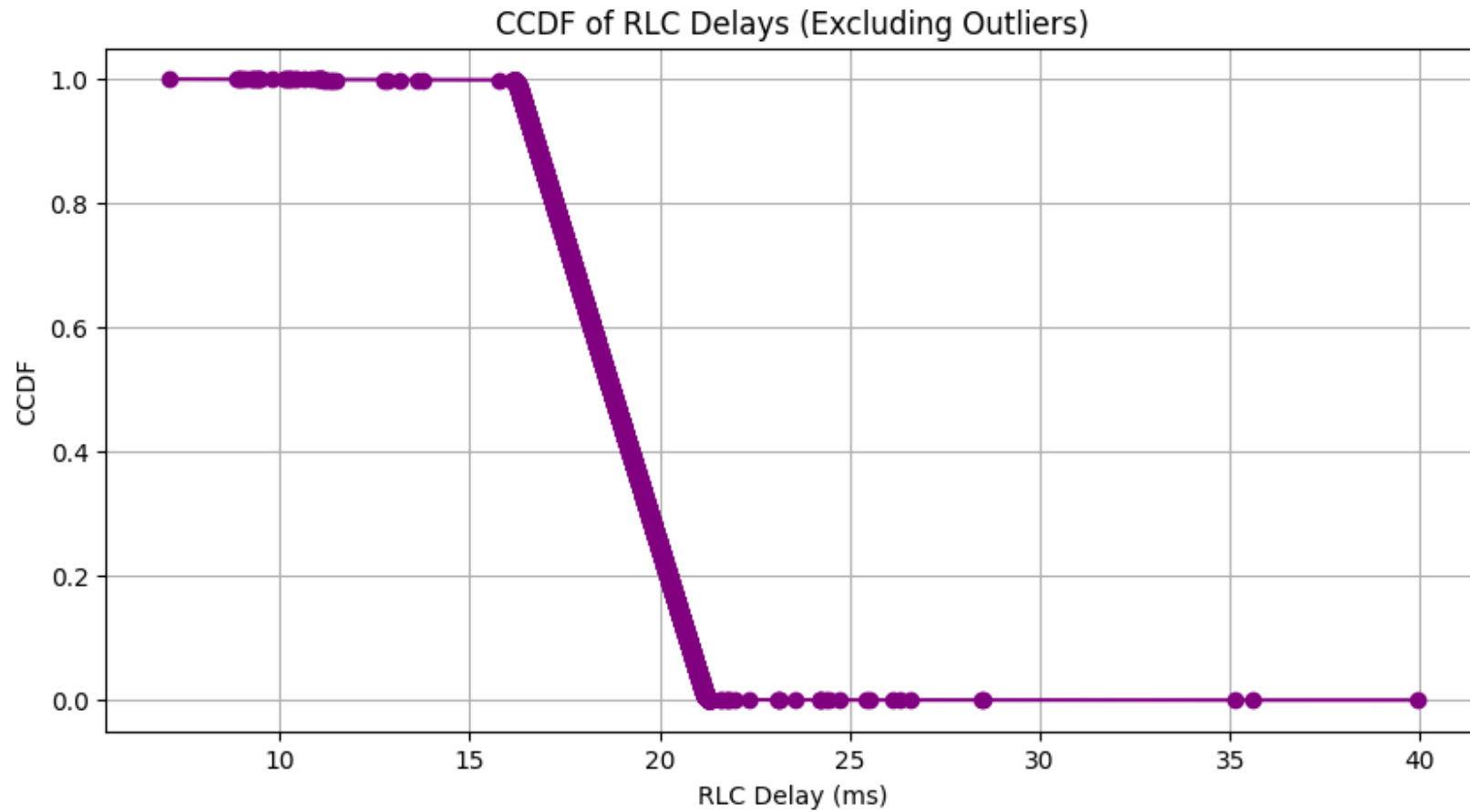
99.8th Percentile: 21.362 ms

99.9th Percentile: 100.060 ms

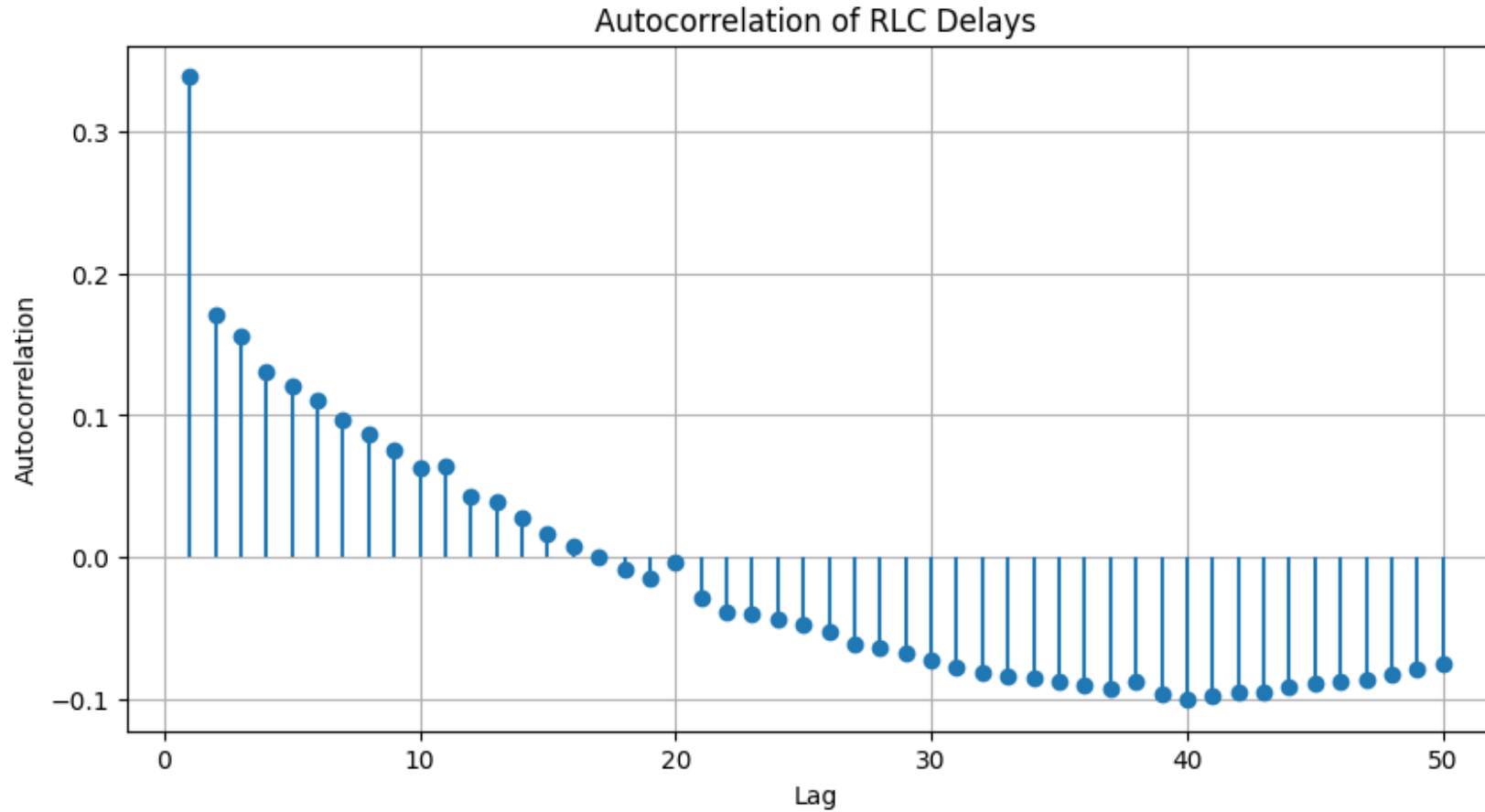
Plot of Outliers in RLC delay ($> 40\text{ms}$)

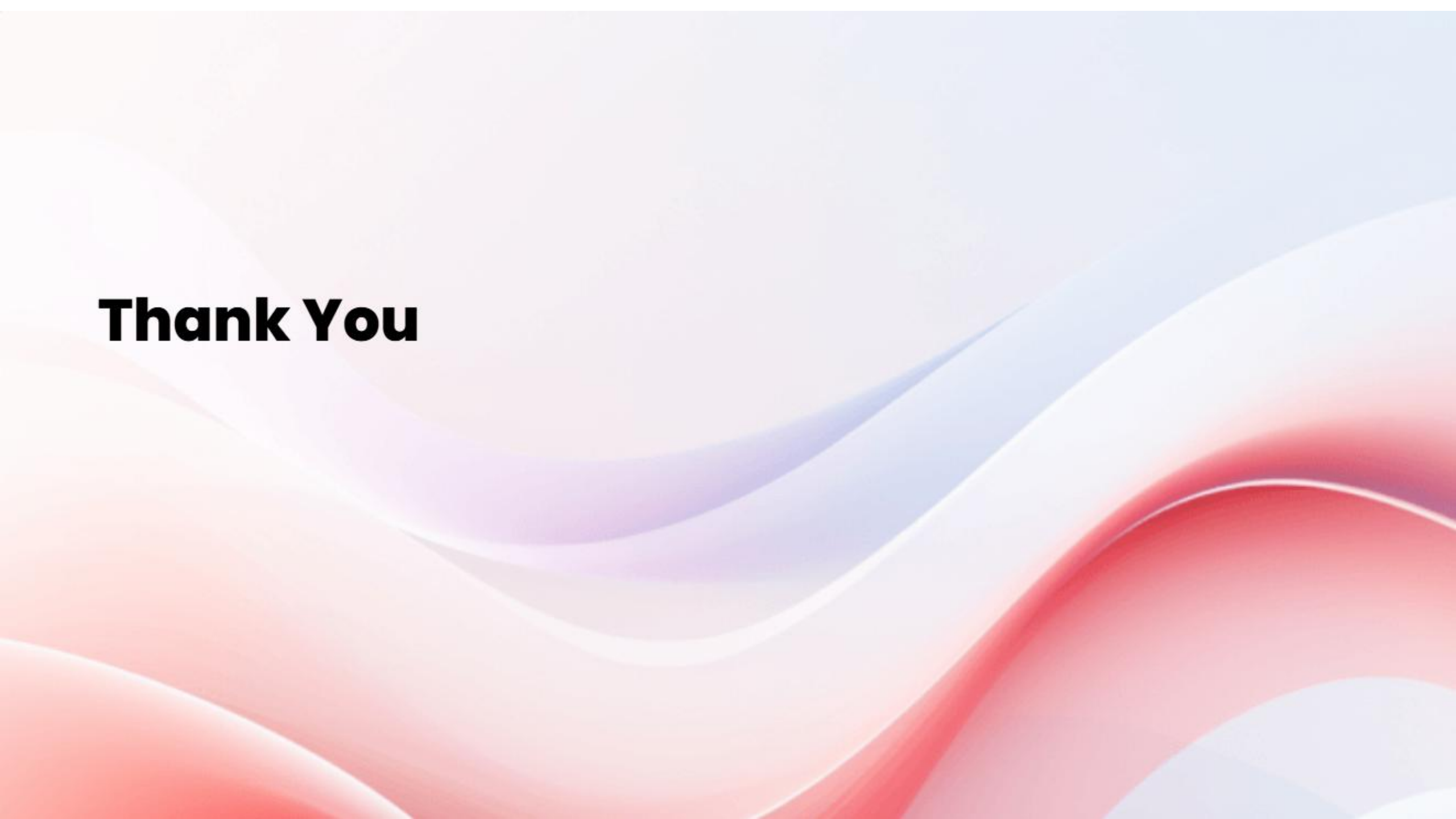


CCDF PLOT OF RLC DELAY(< 40ms)



PLOT OF AUTOCORRELATION OF RLC Delay





Thank You